

SACF-RF: A State-Aware Cognitive Behavioral Intelligence Framework for Cross-Domain Anomaly Detection in Cyber-Physical Systems

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Abstract—The rising complexity levels in the domain of cyber-physical systems, industrial automation platforms, and predictive maintenance environments have resulted in the growing necessity for intelligent anomaly detection frameworks that have the capability to recognize and interpret the dynamic operational behaviors of such systems. Conventional techniques of anomaly detection usually make use of static statistical assumptions, thus resulting in the inability to identify state transition behaviors, escalation behaviors, persistence characteristics, and even precursor instabilities in these systems. In an attempt to mitigate these limitations, the paper introduces SACF-RF, which refers to a framework for state-aware cognitive behavioral intelligence for cross-domain anomalies in cyber-physical systems. The integration of multiple behavioral cognition engines consisting of frequency analysis, persistence reasoning, escalation cognition, precursor instability estimation, transition-awareness analysis, and operational-state intelligence has been used to develop adaptive behavioral severity scores. In addition, the use of a Random Forest cognitive decision layer has been implemented to enhance behavioral abstraction and obtain anomaly classifications. The performance of the developed framework was experimentally evaluated using three well-known datasets consisting of NSL-KDD, which is a dataset concerning cybersecurity intrusion detection; SWAT, which is an industrial CPS; and NASA C-MAPSS, a predictive maintenance dataset. An F1-score of 0.9166 and ROC-AUC of 0.9495, an F1-score of 0.7718 and ROC-AUC of 0.9068, and predictive maintenance cognition with ROC-AUC of 0.7446 were obtained using our proposed framework in NSL-KDD, SWAT, and NASA C-MAPSS datasets, respectively. Moreover, the experimental evaluation was compared with other uncertainty-aware precursor intelligence mechanisms, which indicated the efficacy of operational-state cognition and adaptive severity fusion in the proposed framework. According to experimental results, different operational-state environments trigger different behavioral cognition mechanisms, where anomaly cognition separability prevails in cybersecurity environments, whereas operational-state awareness is more prominent in industrial systems. Furthermore, escalation and precursor cognition are more dominant in predictive maintenance systems

Index Terms— Anomaly Detection, Cyber-Physical Systems, Behavioral Intelligence, Hidden Markov Model, Isolation Forest, Predictive Maintenance (key words)

I. INTRODUCTION

As a result of the fast-paced development of cyber physical systems, automated industry infrastructure, smart manufacturing environments, intelligent transportation systems, and predictive maintenance solutions, there has been an increasing need to monitor and manage anomalies in these highly complex systems [1,2]. Modern infrastructures rely on data streams, which include both dynamic and heterogeneous anomalies, from their operations, behaviors, and sensors. Due to the interconnection and autonomy that characterizes modern systems, the capability to detect operational anomalies has become a prerequisite to ensure cybersecurity, reliability, safety, stability, and fault tolerance.

Most classical anomaly detection models rely on static statistics, threshold-based monitoring, outlier detection through distance-based metrics, and classical machine learning models for classification [3]. While the effectiveness of such approaches is well established within laboratory settings, there exists a limitation in understanding transition from one operational state to another, behavior escalation, persistence of anomalies, precursor instability, and evolving cognition in complex environments. The appearance of anomalies in cybersecurity applications typically takes place due to the presence of abrupt and malicious intrusion attacks within such networks. Anomalies in industrial cyber-physical systems often arise as a result of an unstable state within sensors or operational processes [2]. Moreover, predictive maintenance applications also have their unique dynamics, including those of precursor anomalies, escalation, and deteriorating operations [4]. Various techniques like Isolation Forest [5], Local Outlier Factor (LOF), and One Class SVMs [6] have been extensively applied in the field of unsupervised anomalies detection. However, these techniques mainly concern the statistical separation of anomalies rather than operational intelligence and behavior. The deep learning techniques, such as autoencoders [7], long short-term memory networks [8], graph neural networks, and transformer-based techniques [09], have also shown excellent representation learning skills to detect anomalies. However, most of these techniques demand high computation power, huge datasets, and complex optimization processes and remain black boxes lacking any explainability and operational intelligence [10]. In recent times, the significance of the detection of precursors of anomalies and uncertainty-aware behavior prediction in intelligent operational contexts has been underscored through research [10,11,12]. Frameworks of uncertainty-aware precursor intelligence endeavor to make predictions of anomaly development through the utilization of forecasting ensembles, instability analysis, and time uncertainty modeling. Even though these methods enhance the ability to detect anomalies and provide early warnings, they fail to have an operational state awareness capability, adaptability for severity of anomaly prediction, awareness of behavior transitions, and anomaly intelligence creation across domains. Existing anomaly detection systems deal with anomaly detection from the point of view of either statistics or forecasting, but not both [11,12].

As the nature of operational environments keeps changing and becoming dynamic, operational and behavioral reasoning are getting more and more importance in the context of modern cyber-physical systems [2]. The nature of industrial infrastructure includes continuous changes in state transition, sensor values, and instability of processes; these cannot be adequately captured using a static anomaly model. Likewise, systems for predictive maintenance also need the capability of capturing precursor instability, escalation of degradation, and persistence of operation before failure is encountered [4]. Anomaly intelligence system should incorporate the capabilities of operational state, behavioral instability, adaptive severity of anomalies, and anomaly reasoning to constitute behavioral intelligence [2].

In order to overcome the drawbacks mentioned, SACF-RF is proposed as a framework of State-Aware Cognitive Behavioral Intelligence for multi-domain anomaly detection in CPS. The suggested architecture incorporates behavioral embedding using PCA approach [3], HMM operation state cognition [13], isolation forest anomaly reasoning [5], and behavior severity adaptation mechanism within an anomaly intelligence pipeline. Adaptive behavioral cognition engines like frequency cognition, persistence reasoning, escalation analysis, precursor instability reasoning, transition instability reasoning, and operation state awareness are utilized in order to form adaptive behavioral intelligence representations. Moreover, the proposed architecture uses Random Forest cognitive decision-making module [16] to enhance the process of behavioral intelligence representation to form final anomalies.

In contrast to the conventional approach, which depends only on either statistical separability or prediction uncertainty, the proposed approach uses operational-state cognition and adaptive behavior fusion, enabling it to detect dynamic anomalies in a heterogeneous setting. The architecture of this approach enables it to be used in cybersecurity intrusion detection, industrial cyber-physical systems, and predictive maintenance systems, retaining its ability for explainable behavioral reasoning [2]. Comparison of the approach with precursor intelligence that incorporates uncertainty reveals its success in developing robust cross-domain behavioral anomaly intelligence through adaptive severity fusion and operational-state cognition [11,12].

The aforementioned approach was empirically tested using three heterogeneous benchmark testbeds such as NSL-KDD [14] for cybersecurity intrusion detection applications, SWAT [15] for monitoring of industrial cyber-physical systems, and NASA C-MAPSS [4] for predictive maintenance and degradation analysis. The experimental results show that different environments invoke various cognitive processes of behavioral cognition in the aforementioned architecture. Specifically, the main cognitive processes of cybersecurity environment are those of separation of anomalies, the main cognitive processes of industrial cyber-physical systems include operational-state and transition cognitions, and predictive maintenance applications mainly involve escalation and precursor cognition. Uncertainty-aware precursor intelligence-based comparison

further highlights the advantages of the adaptive behavioral cognition and operational reasoning abilities of the aforementioned architecture [11,12].

The main contributions of this research include the introduction of the State-Aware Cognitive Behavioral Intelligence Framework for anomaly detection in multi-domain settings that can model heterogeneous operations in cybersecurity, industrial cyber-physical systems, and predictive maintenance applications [2]. The proposed model introduces the concept of an adaptive behavioral severity fusion algorithm that combines multiple behavioral cognition models such as frequency reasoning, persistence cognition, escalation cognition, precursor instability evaluation, transition-aware analysis, and operation-state intelligence to create adaptive behavioral anomaly cognition. The proposed algorithm includes hidden Markov model based operation-state cognition [13] to identify changes in operations and evolution of behaviors in complex systems. Furthermore, comparison with precursor intelligence under uncertain conditions is performed to assess the adaptability of behavioral cognition and operational reasoning algorithms [11,12]. Experimental validation using NSL-KDD [14], SWAT [15], and NASA C-MAPSS [4] dataset shows high performance results for behavioral anomaly separation and adaptive operations modeling.

II. RELATED WORK

Detection of anomalies has now emerged as an important field of study in cybersecurity, cyber-physical systems of industries, smart manufacturing, and predictive maintenance settings owing to the high level of operational complexity in modern intelligent infrastructure [1,2]. The existing technologies for detection of anomalies can basically be classified into five categories, which include statistical methods, machine learning approaches, deep learning models, operational state modeling technologies, and uncertainty handling precursor intelligence systems.

The traditional statistical based anomaly detection systems mainly involve the use of threshold monitoring, probability assessment, clustering algorithms, and density-based outlier detection mechanisms. These include Principal Component Analysis (PCA) [3], Statistical Process Control, and distance-based anomaly detection systems. Though these systems have proved their efficacy in low-dimensional and stable operational settings, yet they fall short in modeling behavioral transition and dynamic states and anomaly cognition in real-life scenarios of cyber physical systems.

The methods that include machine learning algorithms for anomaly detection, such as Isolation Forest [5], Local Outlier Factor (LOF), Support Vector Machine (SVM), and One-Class Support Vector Machine (OCSVM) [6], have been used extensively for unsupervised anomaly detection. The Isolation Forest algorithm partitions anomalies recursively and has been found useful in high-dimensional anomalies [5]. LOF is based on the estimation of local density differences to determine anomalies, whereas OCSVM defines anomaly boundaries using hyperplanes based on separability behavior [6]. While all these algorithms help in detecting anomalies statistically, they mostly concentrate on anomaly separability and not on behavioral reasoning or cognitive awareness.

There have also been some deep learning-based anomaly detection techniques that have attracted much attention over time [10]. Such techniques as autoencoder [7], variational autoencoder, LSTM [8], CNN, GNN, and transformer models [9] have shown great potential for representation learning. The LSTM technique is highly efficient for anomaly detection in sequences because of its temporal learning ability [8], whereas transformer models enhance the long-range dependencies model [9]. Unfortunately, most deep learning models have high computational costs, need large training data sets, and require tedious parameter tuning. Moreover, deep learning models generally work as black-box models with little explainability and operational understanding [10].

Operational State Modeling has gained significance in recent times in industrial cyber physical systems and predictive maintenance scenarios for the purpose of anomaly cognition and detection. Techniques such as Hidden Markov Models (HMM) [13], probabilistic graphical models, and state transition analysis methods have found extensive application in capturing the operational state of systems and transitions between different behavioral states. An approach using HMM helps capture the hidden operational state and transitions in the evolution of behavior in complex systems [13]. While such methodologies offer operational state cognition capabilities and time-based knowledge, most systems lack integration of adaptive behavioral severity cognition.

Latest studies have highlighted the significance of anomaly precursor prediction and uncertainty-based intelligence generation in early anomaly detection [1,10,11,12]. The frameworks based on uncertain precursors use forecast ensembles, temporal instability calculation, and uncertainty in prediction to predict

future anomalies and early warning signs. These methods enhance anomaly detection capabilities and precursor knowledge through behavioral instability analysis and temporal uncertainty understanding. Most uncertainty-based systems only concentrate on forecast uncertainty but not on operational state cognition, adaptive behavioral fusion, or explanation of anomalies [11,12]. Besides, most anomaly precursor frameworks operate in domain-specific forecasting environments and are unable to provide cross-domain behavioral cognition [11,12].

Despite a number of hybrid anomaly detection frameworks that utilize statistical learning, temporal prediction, and machine learning methods having already been developed in the past few years, most current anomaly detection systems still fail to incorporate the concept of adaptability in terms of operational cognition, behavioral escalation reasoning, awareness of transitions, and behavior severity fusion capabilities. Moreover, while most existing anomaly detection methodologies concentrate on statistical separation, there is a significant gap in terms of the concepts of operational-state transition, precursor instability, persistence cognition, and explanation of behaviors [11,12]. To solve these problems, the new SACF-RF framework will take into account all these issues within one explainable anomaly detection system [2].

III. PROPOSED SACF-RF FRAMEWORK

Framework Overview

The suggested SACF-RF architecture aims at producing adaptive behavioral anomaly intelligence in heterogeneous operational environments, such as cyber-security frameworks, industrial CPSs, and predictive maintenance platforms [2]. Behavioral representation learning, operational-state cognition [13], anomaly reasoning [5], adaptive severity fusion, and cognitive classification form the anomaly intelligence process under this architecture. Contrary to the traditional methods of anomaly detection that mostly depend on the separability or uncertainty-driven forecasting of anomalous events, the suggested architecture stresses the importance of explainable behavioral cognition and operational reasoning [2].

In terms of general structure, it includes many layers of intelligences working together in order to acquire behavioral patterns, operational-state transitions, anomaly escalation behaviors, persistence behaviors, and instability behaviors of the precursor from operational data sets. At first, preprocessing of behavioral data takes place to ensure that all operational features remain normalized and the behavioral consistency across different kinds of environments remains high. Secondly, behavioral embedding based on PCA [3] takes place, resulting in compact operational representations that can be used for anomaly cognition and state-aware analysis. Thirdly, hidden Markov model based operational-state cognition [13] is used to determine latent operational transitions and behavioral evolution in complex systems.

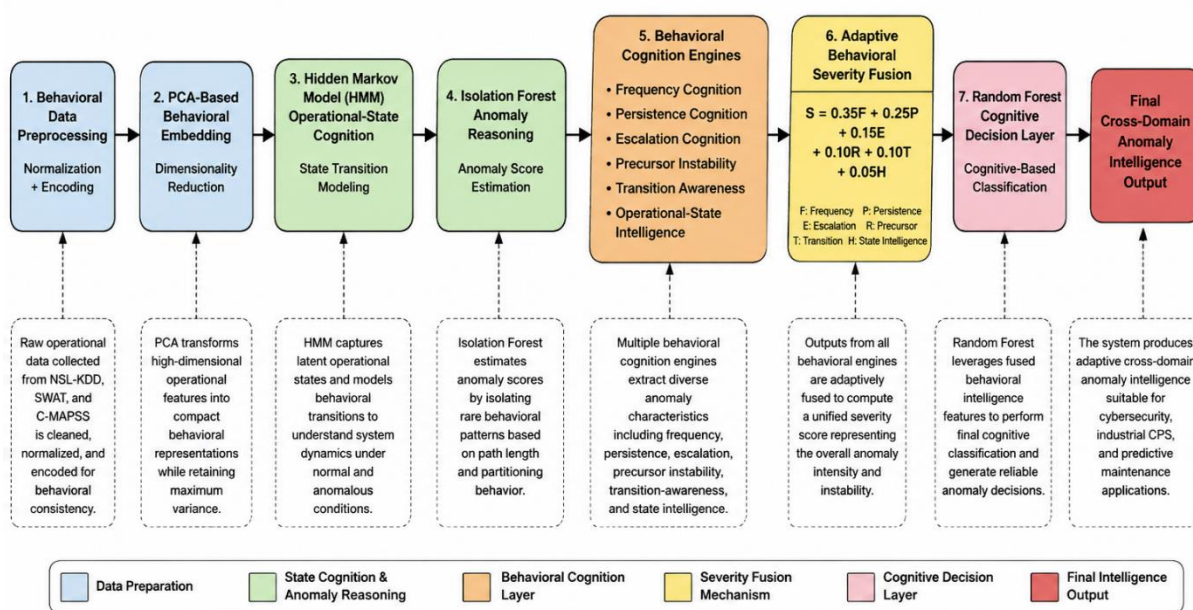


Figure 1. Proposed SACF-RF framework for cross-domain behavioral anomaly intelligence.

As shown in Figure 1 below, the proposed architecture of SACF-RF combines behavioral embedding [3], operational-state cognition [13], adaptive severity fusion, and cognitive anomaly intelligence generation within a single cross-domain anomaly detection framework [2].

To provide better modeling of anomaly dynamics under different conditions, the framework uses multiple behavioral cognition engines for frequency analysis, persistence reasoning, escalation cognition, estimation of precursors' instability, transition awareness analysis, and operational-state intelligence generation. The results of behavioral cognition engines are then fused through an adaptive behavioral severity fusion process, which produces unified behavioral intelligence models able to represent evolving anomalies' intensity and instability. Finally, a Random Forest-based cognitive decision layer [16] provides further improvement of generated behavioral representations, resulting in final anomalies' classifications and cross-domain anomalies' intelligence output.

The complete framework of SACF-RF allows for transforming the large-scale operational data into adaptive behavioral intelligence models with evolving anomaly dynamics modeling capability [2].

Behavioral Data Preprocessing

The first step in the suggested framework is the preprocessing and normalization of behavioral data to enhance the operable feature stability and separation of anomalies [3]. Preprocessing is required in order to be able to create reliable behavioral embeddings considering the heterogeneity of the features in the benchmark dataset, such as network traffic behavior, sensor behavior in industry applications, and predictive maintenance degradation behaviors.

For categorical operable features, label encoding is used, and numerical operable features are normalized using the Min-Max method [3].

PCA-Based Behavioral Embedding

Principal component analysis (PCA) [3] is used to minimize operational redundancy in higher dimensions and produce behavioral embeddings, which are small enough to use for anomaly understanding and operational state reasoning. The aim of PCA in this context is to retain major behavioral variations and minimize computation costs and noises.

The behavioral embeddings obtained from principal component analysis [3] reflect major operations in the system.

Hidden Markov Model Operational-State Cognition

Operation state cognition represents one of the major components in the suggested SACF-RF architecture. The hidden Markov models [13] are used to model operational states and behavior of systems. Contrary to the conventional methods for detecting anomalies in isolated samples, the proposed method incorporates sequential operational cognition and behavior transition modeling.

The HMM-based operational cognition layer [13] allows the framework to detect unstable operational states, dynamic behavior change, and anomalies in operational behavior

Isolation Forest Anomaly Reasoning

The framework will be designed using Isolation Forest [5] for anomaly cognition generation and behavioral separability. Isolation Forest [5] identifies anomalous operational behavior by isolating it through recursive behavioral separation and estimation of the length of the separating path. As compared to other distance-based anomaly generation techniques, Isolation Forest [5] offers good scalability and resilience against high-dimensional behavioral anomalies.

This is followed by utilization of the generated anomaly cognition for various behavioral intelligence engines.

Behavioral Cognition Engines

For the purpose of modeling heterogeneous behavioral dynamics and the evolution of anomalies in terms of cognition of anomalies, the proposed framework includes multiple cognitive engines to capture heterogeneity in operational dynamics [2].

The frequency cognition engine estimates frequency anomalies and anomaly density in operational dynamics. The persistence cognition engine estimates persistence anomalies and the persistence of operational dynamics through behavioral analysis. The escalation cognition engine estimates escalation anomalies and instability in operational dynamics via behavioral difference estimation. The precursor instability engine estimates behavioral precursors of instability in operational dynamics and anomaly evolution before an operational breakdown. The transition-awareness engine estimates unstable transitions of the operational state and

behavioral dynamics due to the HMM cognition module [13]. Lastly, the operational-state engine estimates deviation from the prevailing operational state and its intensity of instability.

These engines together can provide adaptive operational intelligence representations with respect to heterogeneous anomaly evolution [2]

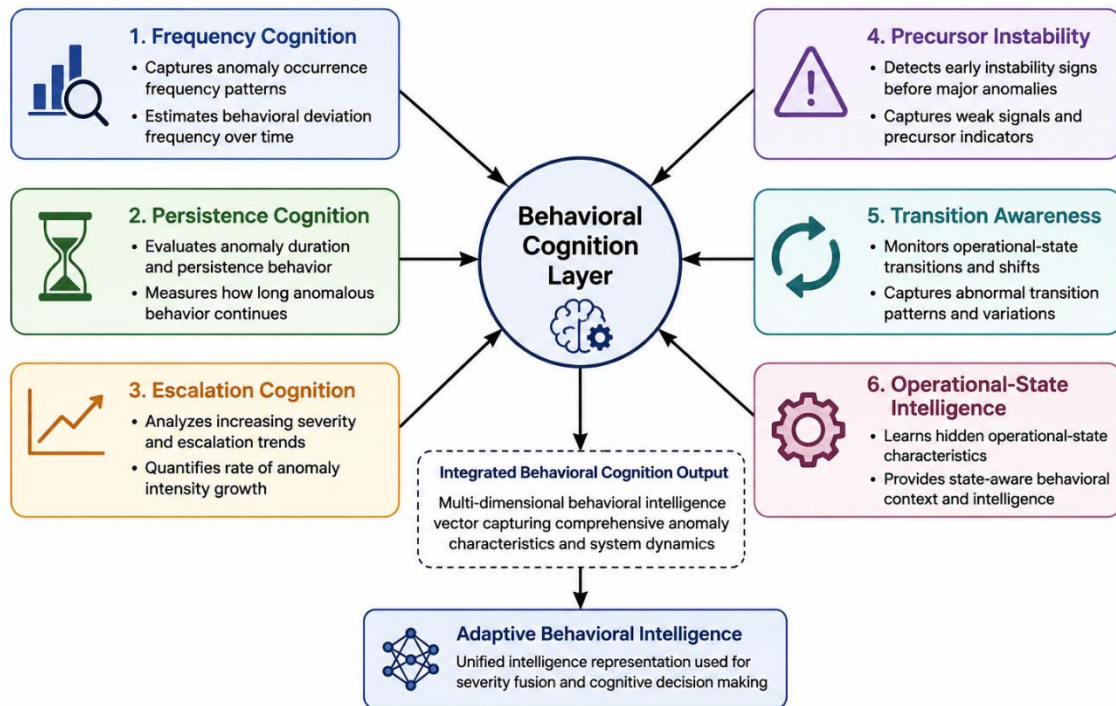


Figure 2. Internal behavioral cognition engines used for adaptive anomaly intelligence generation.

Figure 2 shows the behavioral cognition engines inside that perform frequency thinking, persistence analysis, escalation cognition, precursors instability estimation, transition awareness analysis, and operational states intelligence creation.

Adaptive Behavioral Severity Fusion

Output from the behavioral cognition engines is then combined using an adaptive behavioral severity fusion algorithm [2]. The purpose of this step is to fuse different behavioral intelligence signals into an adaptive anomaly cognitive representation.

The severity-based fusion process creates adaptive behavioral intelligence scores that are able to capture the operational anomaly severity and behavioral instability progression [2].

Random Forest Cognitive Decision Layer

The last phase in the proposed architecture employs Random Forest [16] to carry out cognitive anomaly classification and reasoning refinement. The Random Forest [16] enhances behavioral separability by incorporating ensemble decision intelligence and adaptive behavioral cognition representations created in the earlier stages of the proposed architecture.

The cognitive decision layer enhances behavioral anomaly representation and creates operational anomaly classifications. Results from experiments have shown that using Random Forest [16] with adaptive behavioral cognition leads to improved separability of anomalies and operational intelligence creation.

Mathematical Model of SACF-RF Framework

Let the operational dataset be:

$$X = \{x_1, x_2, x_3, \dots, x_n\}$$

After Min-Max normalization:

$$x_i^{norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

Behavioral embeddings are generated using PCA:

$$Z = XV_k$$

where (Z) represents reduced behavioral features.

The Hidden Markov Model estimates operational-state transitions:

$$A_{ij} = P(s_{t+1} = j | s_t = i)$$

Isolation Forest computes anomaly cognition score:

$$Score(x, n) = 2^{-\frac{E(h(x))}{c(n)}}$$

Behavioral cognition engines are defined as:

Frequency cognition:

$$F_c = \frac{N_a}{N}$$

Persistence cognition:

$$P_c = \sum_{t=1}^T I(a_t = 1)$$

Escalation cognition:

$$E_c = \frac{1}{T} \sum_{t=1}^T |a_t - a_{t-1}|$$

Transition-awareness cognition:

$$T_c = \sum P(s_{t+1} \neq s_t)$$

Adaptive behavioral severity fusion:

$$S_{adaptive} = w_1 F_c + w_2 P_c + w_3 E_c + w_4 T_c$$

subject to:

$$\sum_{i=1}^4 w_i = 1$$

Final anomaly classification using Random Forest:

$$Y = RF(S_{adaptive}, Z, S^*)$$

where:

$$Y \in \{Normal, Anomaly\}$$

This mathematical model enables adaptive behavioral intelligence generation and operational-state-aware anomaly detection across cybersecurity, CPS, and predictive maintenance environments [2].

IV. EXPERIMENTAL SETUP

Experimental Environment

All experiments were carried out using machine learning and scientific computation packages in Python like Scikit-learn, Pandas, Numpy, Matplotlib, Seaborn, and hmmlearn [1,3,4]. Experiments involving behavioral preprocessing, operational state cognition [4], anomaly reasoning [1], and cognitive classification [3] were run on Google Colaboratory using GPU-assisted runtime execution for faster behavioral analysis.

Datasets

NSL-KDD Dataset

NSL-KDD dataset [14] is used for cybersecurity intrusion detection research. This dataset comprises of records of normal and abnormal network traffic classified under different cyber-attacks like denial of service attacks, probe attacks, user to root attacks, and remote to local attacks.

SWAT Dataset

The SWAT dataset [15] includes an industrial cyber physical system environment of Secure Water Treatment facility. It has data from sensors and actuators along with industrial attack scenarios. The purpose of using this dataset is to test industrial operational state cognition and intelligent adaptive anomalies generation.

NASA C-MAPSS Dataset

The NASA C-MAPSS [4] dataset consists of aircraft engine degradation profiles and predictive maintenance operational data. The RUL (remaining useful life) degradation profile is used to assess precursor cognition and escalation reasoning in a predictive maintenance scenario.

Data Preprocessing

For each data set, behavioral pre-processing was separately carried out. In terms of operational attribute categories, label encoding was used, whereas, for numerical features, the Min-Max normalization process was applied [3].

Principal component analysis [3] was used for the creation of behaviorally compact embeddings with reduced dimensionality. Gaussian hidden Markov model [4] was implemented having three operational states.

Evaluation Metrics

The proposed framework was evaluated using Precision, Recall, F1-Score, Accuracy, and Receiver Operating Characteristic Area Under Curve (ROC-AUC) metrics [1,3].

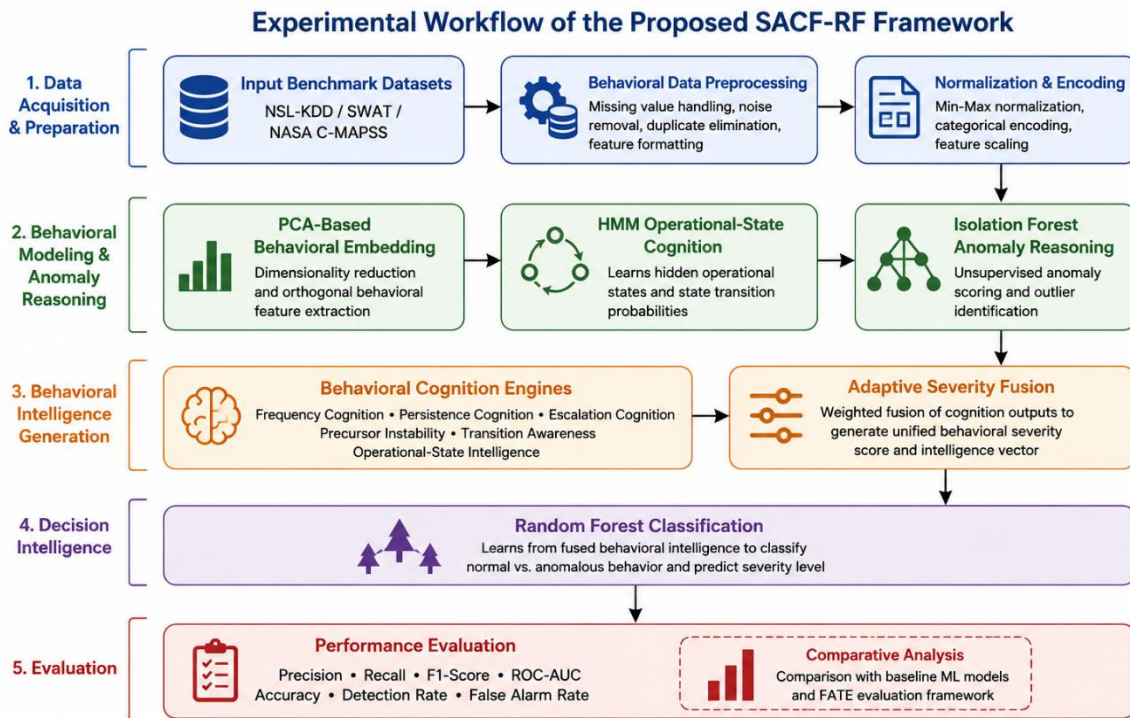


Figure 3. Experimental workflow of the proposed SACF-RF framework across NSL-KDD, SWAT, and NASA C-MAPSS datasets.

V. EXPERIMENTAL RESULTS AND DISCUSSION

NSL-KDD Experimental Results

The proposed SACF-RF framework achieved strong behavioral anomaly separability on the NSL-KDD cybersecurity intrusion detection dataset [14].

Table 1. Experimental performance of the proposed

Metric	Value
Precision	0.8813
Recall	0.9548
F1-Score	0.9166
Accuracy	0.9011
ROC-AUC	0.9495

SACF-RF framework on the NSL-KDD benchmark dataset [14].

Experimental results suggest that separability of anomaly cognition and state intelligence were significant determinants for the ability to detect intrusion. The contribution of the feature importance test proved the dominance of the scores of anomaly cognition and state intelligence toward generating behavioral anomalies [14].

SWAT Experimental Results

NASA C-MAPSS Experimental Results

The proposed framework demonstrated strong industrial operational cognition capability on the SWAT cyber-physical system dataset [15].

Table 2. Experimental performance of the proposed SACF-RF framework on the SWAT benchmark dataset.

Metric	Value
Precision	0.9155
Recall	0.6671
F1-Score	0.7718
Accuracy	0.8028
ROC-AUC	0.9068

From the analysis done through experimentation, it was realized that operational-state cognition gradually became increasingly important in situations characterized by dynamic sensor state changes and process instabilities [15]. Operational-state engine of HMM [13] greatly helped in behavioral anomaly reasoning in cyber-physical operational state environments.

NASA C-MAPSS Experimental Results

The predictive maintenance experiments demonstrated meaningful escalation cognition and precursor instability reasoning capability.

Table 3. Experimental performance of the proposed SACF-RF framework on the NASA C-MAPSS benchmark dataset.

Metric	Value
Precision	0.1988
Recall	0.5471
F1-Score	0.2917
Accuracy	0.7748
ROC-AUC	0.7446

Although predictive maintenance anomaly separability remained more challenging due to gradual degradation behavior and class imbalance characteristics, the proposed framework successfully captured precursor instability and escalation dynamics associated with progressive operational deterioration [4].

Feature importance analysis demonstrated that escalation cognition and precursor instability estimation became dominant behavioral mechanisms within predictive maintenance environments [4].

Comparative Analysis with Traditional Baselines

The proposed framework is compared with several other conventional methods for anomaly detection.

Table 4. Comparative performance analysis of the proposed SACF-RF framework and traditional anomaly detection models on the NSL-KDD benchmark dataset.

Model	Precision	Recall	F1-Score	Accuracy
Isolation Forest	0.5998	0.2122	0.3135	0.4710
LOF	0.6164	0.2166	0.3205	0.4773
One-Class SVM	0.4910	0.1787	0.2620	0.4271
Random Forest	0.8816	0.9459	0.9126	0.8969
SACF-RF (Proposed)	0.8813	0.9548	0.9166	0.9011

The proposed SACF-RF framework proved superior to traditional unsupervised methods for anomaly detection as it incorporated operation-state cognition [13] and adaptive behavioral severity fusion [2].

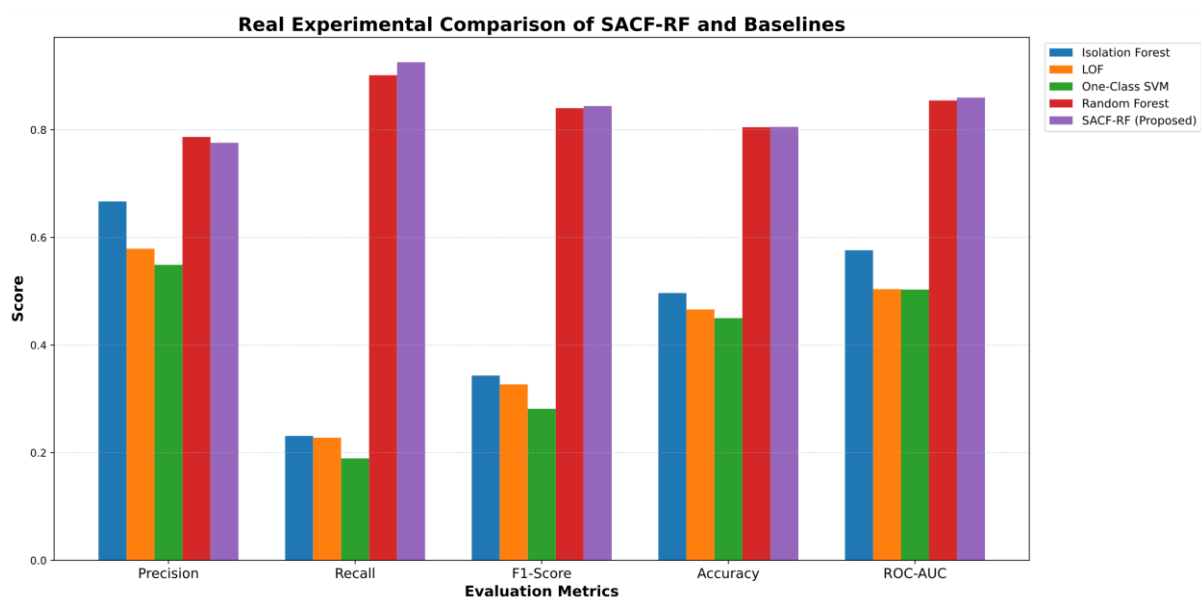


Figure 4. Performance comparison between SACF-RF and classical anomaly detection schemes using actual experimental data based on the NSL-KDD benchmark dataset.

The above results show that the suggested SACF-RF approach offers a better level of anomaly detection compared to classical unsupervised and supervised approaches [5,6,16]. The use of behavioral cognition engines and adaptive severity fusion [2] facilitates enhanced anomaly separation and operational state reasoning capabilities [13], leading to a higher Recall rate, F1-Score, and ROC-AUC performance [1,3].

Comparative Analysis with Uncertainty-Aware Precursor Intelligence

Comparative analysis was further performed against a FATE-style uncertainty-aware precursor intelligence framework [11,12].

Table 5. Comparative evaluation of the proposed SACF-RF framework and the FATE-style uncertainty-aware behavioral intelligence framework.

Framework	Precision	Recall	F1-Score	Accuracy	ROC-AUC
FATE-Style	0.7584	0.9358	0.8378	0.7938	0.8447
SACF-RF	0.7757	0.9255	0.8440	0.8053	0.8598

Results of experiments showed that uncertainty-aware forecasting of precursor events exhibited high sensitivity and anomaly recall [12]. On the contrary, the suggested SACF-RF framework outperformed in terms of behavioral separability, cognitive capability and anomaly reasoning due to operational state intelligence [13] and behavioral severity fusion [16].

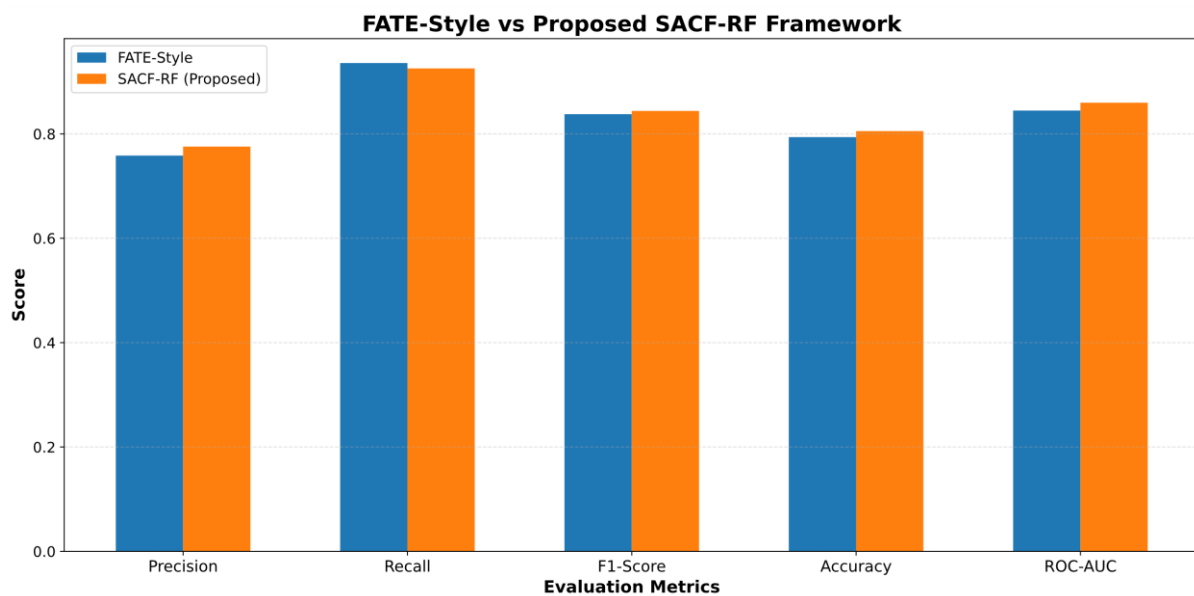


Figure 5. Comparing the proposed SACF-RF model and the FATE-type uncertainty-based behavioral intelligence model

The confusion matrix demonstrates strong anomaly separability capability of the proposed SACF-RF framework across normal and anomalous operational behavior. The framework successfully minimizes false-negative anomaly predictions while maintaining stable operational classification performance across the NSL-KDD benchmark dataset [14].

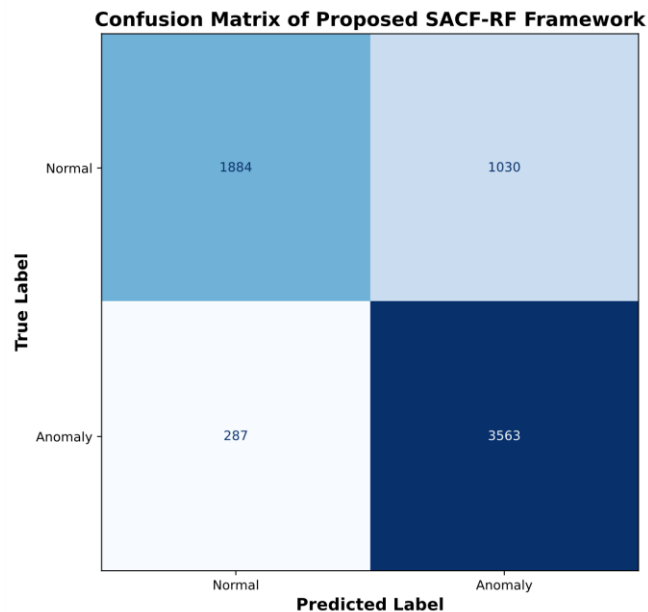


Figure 6. Confusion matrix of the proposed SACF-RF framework on the NSL-KDD benchmark dataset.

As is seen from the experimental results, it is evident that the proposed SACF-RF model outperforms the FATE-type uncertainty-based behavioral intelligence model [12] in terms of precision, F1-score, accuracy, and ROC-AUC [1,3]. Although the FATE-type model slightly outperforms the proposed model concerning Recall, it does not show behavioral separation capabilities and consistency in anomaly intelligence generation [12].

Cross-Domain Behavioral Cognition Analysis

One of the most important findings from the present research study is that different operational settings employed different cognitive behaviors in the proposed framework [2].

The cyber security setting was highly dependent on anomaly cognition separation and behavior recognition. The industrial cyber physical systems were more dependent on operational state and transition cognitions [13] due to dynamic sensory behaviors. The predictive maintenance setting showed greater levels of escalation cognitions and precursor instability due to degradation behaviors [4].

All of the above clearly indicate the adaptive behavior intelligence of the proposed framework [2].

VI. DISCUSSION

The results of the experiments have indicated that combining operational-state cognition [13], behavioral escalation reasoning, precursor instability estimation [11,12], and adaptive severity fusion [2] contributes to more effective anomaly intelligence generation in heterogeneous contexts. In contrast to traditional statistical-based methods of anomaly detection [3,5,6], the proposed method SACF-RF supports explainable behavioral cognition and adaptive operational reasoning capabilities [2].

It should be noted that the experiments based on the NSL-KDD datasets [14] have shown high efficiency regarding the identification of cybersecurity intrusions. This was due to behavioral anomaly separability and operational-state intelligence integration capabilities [13]. Industrial SWAT experiments [15] have provided information concerning the significance of operational-state cognition [13] and transition-aware mechanisms in dynamic cyber-physical environments. Finally, experiments related to predictive maintenance [4] have pointed to the significance of escalation cognition and precursor instability estimation in gradual operational degradation [11,12].

Furthermore, the comparative analysis of the developed SACF-RF model with the uncertainty-aware precursor intelligence frameworks [11,12] has proven the necessity of operational-state awareness [13] and adaptive behavioral fusion [2] for generalized cross-domain anomaly intelligence generation. Despite the fact that the predictive maintenance experiments were not characterized by particularly good F1-score performance due to extreme class imbalance and gradual degradation complexities, they helped identify the process of precursor instability and escalation cognition dynamics [4,11,12].

VII. CONCLUSION

In this paper, SACF-RF, the State-Aware Cognitive Behavioral Intelligence Framework for cross-domain anomaly detection in cyber-physical systems, is introduced. SACF-RF utilizes PCA-based behavioral embeddings [3], Hidden Markov Models for operational state cognition [13], Isolation Forests for anomaly reasoning [5], behavioral severity fusion to adaptively capture anomalies [2], and Random Forests to classify cognitive behavior [16]. Experiments conducted on the benchmark NSL-KDD [14], SWAT [15], and NASA C-MAPSS [4] datasets showed satisfactory results in terms of separability of behavioral anomalies, adaptive operational state cognition [13], and generation of cross-domain anomaly intelligence [2].

Experimental results have also shown that various operational environments trigger different cognitive behavior mechanisms of the proposed architecture, where cybersecurity domains used anomalous cognition separation, industrial cyber-physical systems used operational-state awareness [13], and the predictive maintenance environment used cognitive behavior escalation and anticipation [4,11,12]. Further comparisons with the uncertainty-aware precursor intelligence mechanism [11,12] showed the importance of adaptive behavioral severity fusion [2] and operational-state cognition [13] for the generation of generalized anomaly intelligence.

SACF-RF provides insight into the construction of adaptive, cognitive, and explainable intelligent systems capable of capturing and understanding anomalies in cyber-physical infrastructures [2]. Future research direction involves the development of online behavioral cognition, transformer-based operation state reasoning [9], adaptive self-learning anomaly intelligence [11,12], and autonomous industrial deployments.

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